Credit: NASA

Touch screen enabled gloves to browse dank memes in outer space

NEW SPACE AGE – AND SUITS!

Humankind always needs to have a frontier, a goal on the horizon to push and strive towards. Since we have pretty much covered all the Earth based land frontiers in the last century, and deep sea exploration is intrinsically tied to space exploration (it just comes down to the pressure differential), countries are starting to fund their space agencies meaningful amounts again. With the advent of technology within the last two decades, more and more complex electronics have shrunk. They are now small enough to add on newer, better materials with incredible properties being invented. We are on the verge of a new space age and our red neighbour is the first one in our sights. NASA is currently developing a line of spacesuits called the Z Series for use in Mars. It utilises a lot of modern composites to make the suit thinner and lighter while not compromising on strength. (More at: http://dgit.in/suitupnasa)

The NASA suit focuses on maximising astronaut productivity and ease of use while on the Martian surface. But the late 50's which works on the basis of the suit itself applying pressure on the human body and therefore cutting a lot of the bulk from the gas. MIT is developing a biosuit based on this technology, each of which will be laser scanned to custom fit an individual. The tension elements will be placed where the body does not stretch in the natural range of motion and elastic cords where the body does stretch, allowing for pretty intuitive motion range as well as decreasing the energy cost to actually perform an action. Right now, they only have the *Lower Torso Assembly* as a working example but they are confident they can finish building a whole suit mission aboard the GSLV - Mark III sometime in the next five years.

SPACESUITS BUT WITHOUT ALL THE SPACE

Technologies developed for spacesuits are actually used in several different fields. When you think about it, deep sea exploration isn't that different from space exploration and spacesuits inspire better diving suits as well as vice versa. There are also the obvious military uses for enhancing soldier's speed and agility while not compromising on armor (HALO anyone?) and effectively creating a super-soldier of sorts. Something just as vitally important, spacesuit technol-

There are all these badass looking suits and then there is Gemini G4-C

it's still a far cry from the skin tight ones that are popular in science fiction. But worry not! Both SpaceX and Massachusetts Institute of Technology are working on making that a reality. Conventional space suits right now use gas within them to pressurise the wearer. The human body requires a little more than 2.5 PSI for things like lung inflation to function properly. But *Mechanical Counter Pressure* suits or MCP was a technology first theorised in soon. SpaceX is keeping tight wraps on their project but a prototype should be out later this year. And as typical of Elon Musk, we expect that it's going to be pretty amazing as well. India is also currently developing it's own spacesuit in Gujarat although it is unknown whether it's going to be an MCP suit or a gas pressure suit. We currently don't have any resources to send astronauts into space right now but plans are currently being made to have an indigenous ogy is now helping disabled people. Researchers are currently working in a Children's Hospital in Boston, USA to help people such as infants born with cerebral palsy or other motor skill impeding disorders, paralysis and stroke victims with 'sleeves' inspired by spacesuits which will effectively grant them the use of their body parts back . Who knows, maybe we all could be wearing spacesuit inspired clothing in the future as well. Straight outta Sci-Fi town.